

Question 1:

```
himasree@LAPTOP-41COT5AA:~/OSLab$ nano program31.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc program31.c -o program31
himasree@LAPTOP-41COT5AA:~/OSLab$ ./program31
```

```
Parent Process
Child Process
Child PID : 676
Parent PID : 675
Parent PID : 675
Child PID : 676
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
    }
    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
    }
    else
    {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }
}
```

Question 2:

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process ID (PID) : %d\n", getpid());
    printf("Parent Process ID (PPID) : %d\n", getppid());

    return 0;
}
```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ nano program32.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc program32.c -o program32
himasree@LAPTOP-41COT5AA:~/OSLab$ ./program32
Process ID (PID)      : 688
Parent Process ID(PPID) : 368

```

Question 3:

```

#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        printf("Child is Running...\n");

        sleep(5);

        printf("Child Terminated.\n");
        exit(0);
    }
    else if (pid > 0)

```

```

    {
        printf("\nParent Process\n");
        printf("PID : %d\n", getpid());

        printf("Parent is Waiting for Child...\n");

        wait(NULL);

        printf("Child Finished.\n");
        printf("Parent Terminated.\n");
    }
    else
    {
        printf("Fork Failed!\n");
    }

    return 0;
}

```

```

himasree@LAPTOP-41COT5AA:~/OSLab$ nano program33.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc program33.c -o program33
himasree@LAPTOP-41COT5AA:~/OSLab$ ./program33

Parent Process
PID : 695
Parent is Waiting for Child...

Child Process
PID : 696
PPID : 695
Child is Running...
Child Terminated.
Child Finished.
Parent Terminated.

```